Scheduling Assignment

Name _____

Student Number _____

SMART CITIES AND COMMUNITIES Instructor: Dr. Marjan Alavi

Deadline: November 8, 2026 Total: 100 points

Notation. r is release time, C service duration, T period, d absolute deadline, $D = d - r$, f finish time, $R = f - r$, and $L = f - d$. Lateness may be negative; $f = d$ is on time. Use intervals $[s, f)$.

1. (35 points) Traffic light scheduling. Use the slides' idealized single-resource model. Green-service jobs are independent and may be interrupted and resumed without loss. Only one job receives service at a time. First releases are at 0; later releases occur every T seconds, with $D = T$. Ignore switching and clearance time for the calculations.

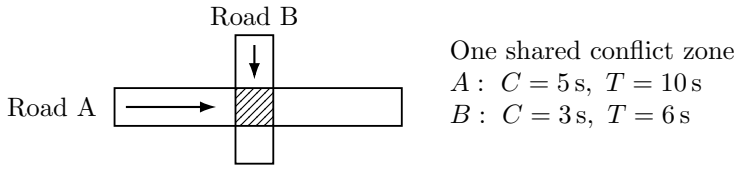


Figure 1. Two conflicting approaches share one resource. This schematic is not an operational signal plan.

Approach	C (seconds)	T (seconds)	D (seconds)
A	5	10	10
B	3	6	6

(a) **(6 points)** List the release and absolute deadline of every job released in $[0, 30)$. Label instances $A_1, A_2, \ldots$ and $B_1, B_2, \ldots$. Calculate $U = \sum_i C_i/T_i$ and $H = \mathrm{LCM}(T_i)$.

(b) **(14 points)** Apply preemptive EDF over $[0, 30)$, checking priorities at each release and completion. Keep the running job on an equal absolute deadline; otherwise break ties alphabetically. Draw the schedule and report f, L, and $L_{\max}$. Are all deadlines met?

(c) **(6 points)** Increase only B's duration to 3.3 seconds. Calculate U and determine whether any schedule can meet this workload indefinitely. For the original workload, explain the effect of adding positive clearance time whenever service switches approaches.

(d) **(9 points)** Real green phases require uninterrupted service and clearance between conflicting approaches. In 80–120 words, explain which analytical assumptions these requirements violate, why the EDF trace alone cannot prove physical feasibility, and whether $L_{\max}$ directly measures vehicle waiting.

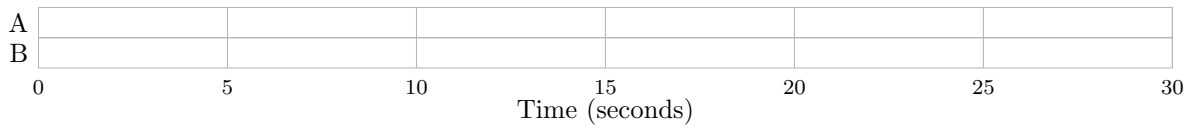


Figure 2. Blank EDF working grid. Mark execution blocks and job identifiers.

Preparation: deck slides 13–15 and 20–27; *Scheduling Algorithms* PDF p. 17 and pp. 24–25.

2. (35 points) Bus scheduling. One community shuttle serves five independent requests, all ready at time 0. The bus serves one request at a time without interruption. Each fixed duration includes boarding, travel, and return to a common dispatch point. Due times refer to completion of the entire service block. Ignore passenger capacity and sequence-dependent travel. All times are minutes after service starts.

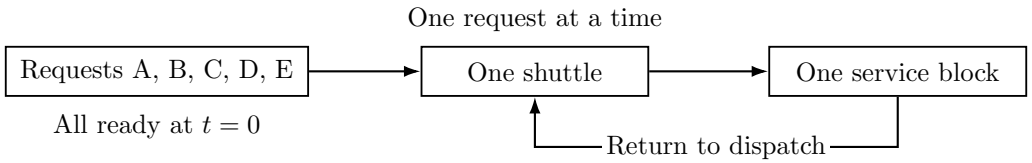


Figure 3. Requests share one shuttle; return to dispatch is included in each block.

Request	Duration C	Completion due time d
A	6	18
B	4	12
C	7	20
D	3	10
E	5	15

(a) **(6 points)** Identify the resource, release pattern, and preemption rule. Explain why EDD minimizes $L_{\max}$ for this model. Distinguish optimizing lateness against service targets from requiring every target to be met.

(b) **(11 points)** Construct the EDD order and Gantt chart. Report each request's start s, finish f, signed lateness L, and tardiness $\max(0, L)$. Calculate $L_{\max}$ and state whether all targets are met.

(c) **(8 points)** Compare EDD with the fixed order A, B, C, D, E. For each, calculate $L_{\max}$ and mean request waiting time, where $W = s - r$. Explain whether EDD generally guarantees minimum mean waiting. This is a request-level mean, not a passenger-weighted mean.

(d) **(5 points)** Show that no order can meet all original due times. Find the smallest common nonnegative extension δ to all due times that makes a feasible one-bus schedule. Recheck $L_{\max}$.

(e) **(5 points)** In 60–100 words, identify two EDD assumptions and explain how this bus model satisfies them. Give one realistic change that breaks an assumption, and state why the theorem would then need rechecking.

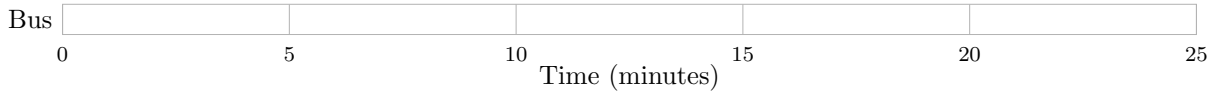


Figure 4. Blank working grid. Label each block with its request identifier.

Preparation: deck slides 8–12 and 23–25; *Scheduling Algorithms* PDF p. 17 and pp. 21–23. The numerical table preserves the deck's bus example.

3. (30 points) Municipal EV charging. Three municipal vehicles remain connected to separate plugs sharing one power module. Exactly one plug is energized at a time. Each job needs C minutes at a fixed power and may pause and resume without loss. Vehicles do not move during switching; switching time is zero. Independent periodic jobs first release at 0, then every T minutes, with $D = T$. Ignore battery tapering and other overhead.

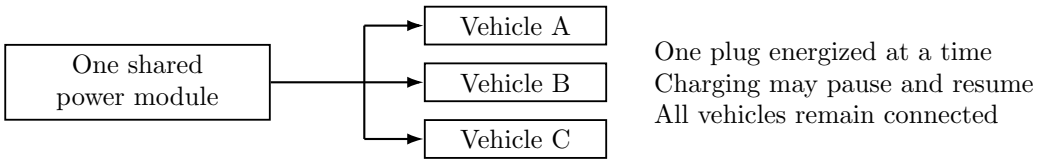


Figure 5. Available connections to one shared power module. The branches do not indicate simultaneous energy delivery.

Vehicle	C (minutes)	T (minutes)	D (minutes)
A	20	60	60
B	10	40	40
C	10	80	80

(a) **(6 points)** Assign RM priorities. Calculate U and the sufficient bound $B = 3(2^{1/3} - 1)$. State what their comparison establishes under the given assumptions.

(b) **(8 points)** Draw the RM schedule over $[0, 120)$, including idle intervals and instance labels. Report f and L for jobs with deadlines at or before 120. Does this trace contain a preemption? Does RM permit preemption in general?

(c) **(10 points)** Design a nonpreemptive cyclic schedule. Use minor cycle $h = \mathrm{GCD}(T_i)$ and major cycle $H = \mathrm{LCM}(T_i)$, following the lecture convention. Tabulate every frame in $[0, H)$, including idle time. Whole jobs may share a frame in a fixed order, but a job cannot cross a frame boundary. Check durations, releases, deadlines, and repeatability.

(d) **(6 points)** A revised three-task workload has $U = 0.81$. Explain what the RM sufficient bound can and cannot tell you. Identify two stated model assumptions whose failure would require another feasibility analysis, and explain the effect of each.

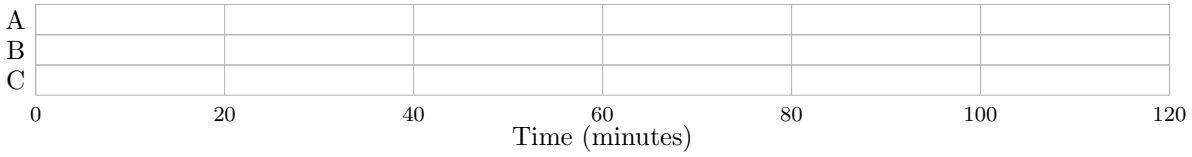


Figure 6. Blank RM working grid for part (b). Prepare a separate frame table for part (c).

Optional discussion, ungraded: suggest a service-fairness metric or consider electricity cost and emissions. These extensions are not required for the assignment or midterm.

Preparation: deck slides 3–4, 16–17, and 29–37; *Scheduling Algorithms* PDF pp. 27–35. The periodic workload scales the deck's collection example by ten minutes per time unit.